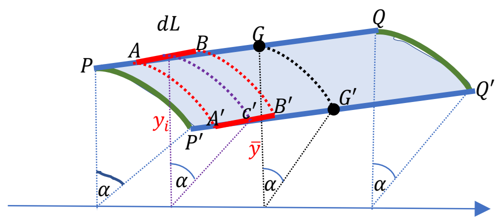
PAPPUS-GULDINUS THEOREMS

Theorem 1

It states that the area of a surface generated by revolving a plane curve about any non-intersecting axis is equal to the product of the length of the curve and distance travelled by its centroid.



Consider any plane curve PQ of length L. Let its centroid G is at a distance $\bar{y}$ from the x-axis. Assume that the curve revolves about the non-intersecting x-axis through some angle α. Then it sweeps an area $PQQ'P'P$ which is shown as shaded. The centroid G also moves to G' moving a distance,

$$GG' = \alpha\bar{y}$$

Let the curve be divided in to elemental lengths and consider one such element AB of length dL whose centre c is at a distance y_i from the x-axis. When the curve revolves about x-axis, the element AB generates a surface $ABB'A'A$. The area of the elemental surface generated,

$$dA = AB \times AA' = AB \times cc' = dL \times \alpha y_i$$

The total area of the surface generated by revolving the plane curve about the x-axis is given by

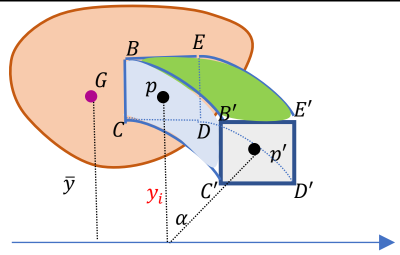
$$A = \int dA = \int dL \times \alpha y_i = \alpha \int y_i dL = \alpha\,\bar{y} \int dL = (\alpha\bar{y}) \times L$$

which is the product of the length of the curve and distance moved by its centroid.

$$\bar{y} = \frac{\int y_i dL}{\int dL}$$

Theorem 2

It states that the volume of a solid generated by revolving a plane surface about any non-intersecting axis is equal to the product of the area of the surface and distance travelled by its centroid.



Consider any plane surface of area A. Let its centroid G is at a distance $\bar{y}$ from the x-axis. Assume that the surface revolves about the non-intersecting x-axis through some angle α. Then it generates a solid.

Let the surface be divided in to elemental areas and consider one such element $BCDE$ of area dA whose centre p is at a distance y_i from the x-axis. When the surface revolves about x-axis, the element $BCDE$ generates a solid $BCDEB'C'D'E'$. The centre p of the element moves a distance

$$\overset{\frown}{pp'} = \alpha\bar{y}$$

The volume of the elemental solid generated,

$$dV = Area\ of\ the\ element \times BB' = dA \times pp' = dA \times \alpha y_i$$

The total volume of the solid generated by revolving the plane surface about the x-axis is given by

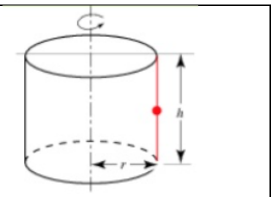
$$V = \int dV = \int dA \times \alpha y_i = \alpha \int y_i dA = \alpha\,\bar{y} \int dA = (\alpha\bar{y}) \times A$$

which is the product of the area of the surface and distance moved by its centroid.

$$\bar{y} = \frac{\int y_i dA}{\int dA}$$

Example 1

Find the curved surface area of a cylinder of base radius r and height h.



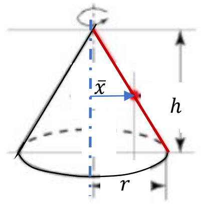
The cylndrical surface can be generated by revolving a vertical straight line of length h (shown in red) about the y-axis through 2π radians (one revolution). The centroid of the line is at its centre and the distance of the centroid from the axis of rotation (y-axis) is $\bar{x} = r$

Area of the cylindrical surface generated by revolving the line with respect to y-axis,

$$A = (\alpha\bar{x})\, L = 2\pi(r)h = 2\pi rh$$

Example 2

Find the curved surface area of a cone of base radius r and height h.



The conical surface can be generated by revolving an inclined straight line of length l (shown in red) about the y-axis through 2π radians (one revolution). The centroid of the line is at its centre and the distance of the centroid from the axis of rotation (y-axis) is $\bar{x} = \frac{r}{2}$

From the similar triangles,

$$\frac{\bar{x}}{r} = \frac{\frac{l}{2}}{l} \ggg \bar{x} = \frac{r}{2}$$

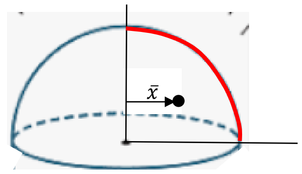
$$\alpha = 2\pi$$

Area of the conical surface generated by revolving the line with respect to y-axis,

$$A = (\alpha\bar{x})\, L = 2\pi\left(\frac{r}{2}\right) l = \pi rl$$

Example 3

Find the curved surface area of a hemi-sphere of base radius r.



The hemi-spherical surface can be generated by revolving a quarter circular arc of radius (shown in red) about the y-axis through 2π radians (one revolution). The centroid of a quarter circular arc is at a distance $\bar{x} = \frac{2r}{\pi}$ from the axis of rotation (y-axis).

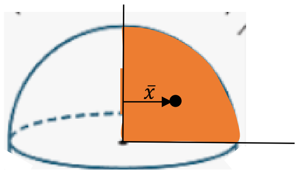
Length of the curve, $L = \frac{\pi r}{2}$

Area of the hemi-spherical surface generated by revolving the quarter circular arc with respect to y-axis,

$$A = (\alpha\bar{x})\, L = 2\pi\left(\frac{2r}{\pi}\right)\frac{\pi r}{2} = 2\pi r^2$$

Example 4

Find the volume of a hemi-sphere of base radius r.



The hemi-sphere can be generated by revolving a quarter circular area of radius r (shown in red) about the y-axis through 2π radians (one revolution). The centroid of the quarter circular arc is at a distance $\bar{x} = \frac{4r}{3\pi}$ from the axis of rotation (y-axis).

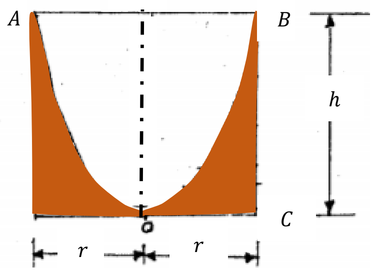
Area of the quarter circle, $A = \frac{\pi r^2}{4}$

Volume of the of the hemi-sphere generated by revolving quarter circular area with respect to y-axis,

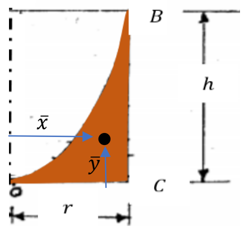
$$V = (\alpha\bar{x})\, A = 2\pi \left(\frac{4r}{3\pi}\right)\frac{\pi r^2}{4} = \frac{2}{3}\pi r^3$$

Example 5

A right circular cylindrical tank containing water spins about its vertical axis at such a speed that the free water surface is a paraboloid AOB. What is the depth of water in the tank when it comes to rest?



The paraboloid is generated by revolving the parabolic spandrel OCB with respect to the axis of rotation. We have already derived the co-ordinates of the parabolic spandrel earlier as $\bar{x} = \frac{3a}{4}$ ⋘⋙ $\bar{y} = \frac{3b}{10}$. Here the paraboloid is obtained by revolving the parabolic spandrel with respect to the y-axis through 2π radians (one revolution).



Area of the parabolic spandrel, $A = \frac{1}{3}rh$

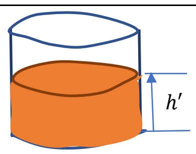
$$\alpha = 2\pi \quad \lll\ggg \quad \bar{x} = \frac{3r}{4}$$

By pappus theorem, the volume of the solid generated by revolving a plane area about a nonintersecting axis is given by the product of the area and distance moved by the centroid.

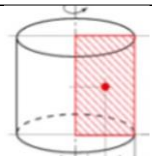
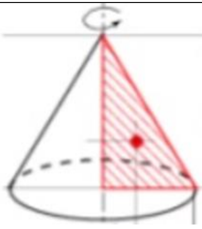
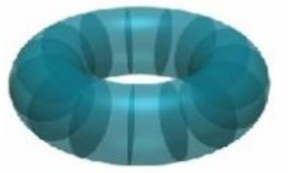
$$V = A(\alpha\bar{x}) = \left(\frac{1}{3}rh\right)\left(2\pi \times \frac{3r}{4}\right) = \frac{1}{2}\pi r^2 h$$

Let h' be the height of water in the cylinder when it comes to rest. The volume of water $V' = \pi r^2 h'$. Assuming no water is lost,

$$V' = V \lll\ggg \quad \pi r^2 h' = \frac{1}{2}\pi r^2 h \quad \lll\ggg \quad h' = \frac{h}{2}$$



Exercise 11

11.1 Find the volume of the cylinder of radius r and height h.	
11.2 Find the volume of the cone of base radius r and height h.	
11.3 Find the volume of the torus shown in the figure which was formed by revolving the circular region bounded by $(x-2)^2 + y^2 = 1$	

Reference

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2. Mechanics for Engineers: Beer & Johnston, McGraw Hill
3. Engineering Mechanics: Merriam & Kraig, Wiley
4. Engineering Mechanics: Hibbeler, Pearson